

Documentation gaps and verification queue

Status: source-reviewed, partial browser hardware check (firmware snapshot: 2026-09-23). The earlier explicit-sync failure was reproduced and resolved on the assembled boards; other entries still need hardware verification. Resolve or confirm them before turning the development guides into a public manual.

Priority	Finding and source	Documentation consequence / next check
High	With Uno Wi-Fi and RTP disabled, the same four-voice, three-unison, all-effects benchmark completed 110,251 DAC writes in five seconds : the requested 22,050/s average. A timing-instrumented run had no dropped catch-up windows, but 2,381 writes began over 100 μ s late and the worst recorded lateness was 1,734 μ s. Disabling matrix redraws made only a small difference, so that change was rejected. Uno audio loop , current test record .	Measure physical DAC intervals and audible output under sustained four-voice play and MIDI traffic. Check RTP note latency, 40/120/240 BPM clock and transport through the Pico. The output circuit is unbuilt, so the sample counter is not a measured audio waveform or proof of evenly spaced 22.05 kHz output.
Medium	The reproduced explicit-sync timeout was resolved by parsing the Uno's fixed six-byte status replies directly on the Pico. The automated release-build sequence stayed connected through explicit sync, Randomize, and patch loads. Pico link parser , Uno receive loop , test record .	Remaining acceptance: measure the Pico RX high voltage, disconnect the return wire to check offline detection, force a lost acknowledgment, and test both boards starting together and either board starting late.
High	Standalone activity is tracked when DIN delivers Note On/Off, CC, pitch bend, or an accepted QuarkWave sound command, or when the Pico marks forwarded RTP sound activity; MIDI clock and transport are excluded. No physical-controller-presence detection exists. Uno external handlers , Pico auto-sync rule .	Run the Uno without the Pico, attach a controller, change the sound, then attach the Pico. Confirm the sound persists until explicit Sync Pico patch to Uno , and that the warning precedes the reset.
Medium	Fresh user slots are now empty, while any existing file—including a legacy <code>P0 ... P7</code> file or unreadable file—is preserved as occupied. Save As requires an explicit destination and confirmation for replacement. Pico storage and save handler , Save As dialog .	On hardware, check blank storage, upgrades with legacy files, full slots, unreadable files, write failures, and reboot after loading a factory or user patch. Confirm save-result messages and displayed selection.

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Medium	The redesigned panel wires its visible buttons to the existing <code>type</code> messages; Randomize sends one <code>randomize</code> request to the Pico and returns <code>patchData</code> . Browser rendering and control messages were checked in headless Chrome; per-control Uno MIDI reception and audio remain unverified. The new VU drawing is source-reviewed and its selection command reached the live Uno, but the moving bars have not been visually accepted on the board. Browser handlers and view switching , Pico dispatch , Uno VU display .	Browser checks passed for 50 control paths, 41 pointer keys, view switching with a held browser note, Randomize, Load, Save As dialog, rejected saves, visualization messages, and Panic message. Test the VU bars during quiet, sustained, and percussive notes; then test Uno response and audio, successful Save/Save As, Commit/Load Commit, and repeat view/keyboard checks after link recovery.
Medium	The Uno top row now has Pico sound, external activity, clock, sustain, arpeggiator, voice-steal, and audio-scheduling indications in addition to the existing link, receive, note, and heartbeat pixels. The source ran during USB-audio tests: the Pico link stayed connected and the owner confirmed the matrix responds to notes, but the new status pixels have not been individually accepted. Uno LED code , LED guide , test record , USB-audio experiment .	Observe the new pixels on the assembled boards, especially five overlapping notes for voice stealing and a temporary forced-delay diagnostic for column 11. Verify the note flash in Status, VU, and Scope.
Medium	DIN and the Pico RTP gateway now accept sound Program Changes and QuarkWave SysEx 01–03, while visualization, reset, and link commands remain Pico-only. Each input has its own clock history, with DIN priority and transport-controlled arpeggiator scheduling. Uno input handlers , Pico gateway .	On hardware, test accepted and rejected commands, 40/120/240 BPM clock, beat alignment, Start/Continue/Stop, DIN priority, two-second failover, and held tempo after clock loss.
Medium	The Pico now permits one RTP session, tracks its held notes and sustain pedal, and releases both when AppleMIDI reports disconnection. The Uno distinguishes browser, DIN, and wireless ownership. Live RTP disconnect diagnostics confirmed note release, preservation of the same browser-held pitch, wireless pedal release, and preservation of a browser-held pedal. Pico release , Uno ownership , test record .	Test DIN overlap after its physical port is built, long silent network loss until AppleMIDI timeout, and multiple-channel RTP note overlap. Use browser Panic if an immediate release is needed before timeout.

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Medium	The Mac and Windows network MIDI setup follows the Pico's session and port plus Apple and rtpMIDI setup instructions. A direct RTP invitation and note passed a two-board smoke check, but the Mac/Windows procedures, Logic-to-Pico MIDI sequencing, Windows 11 compatibility, and Logic external-clock timing have not been accepted on the assembled instrument. The owner confirmed Logic with the separate USB-audio input without specifying whether RTP was used. Pico session , USB-audio result .	Walk through both platform procedures, confirm a note and note release from each DAW, record MIDI and USB audio in Logic, then test tempo/transport and disconnect. Record the exact Windows version and MIDI driver version.
Medium	The owner's photographs show the two USB-powered boards and level-shifter module; the owner confirms the return-wire shift and common ground. Both UART signal wires appear to use separate level-shifter channels; exact channel pairs, LV/HV supply terminals, and Pico RX voltage are unmeasured. The Uno's <code>A0</code> audio stage is not built. The owner also has a TDA1308 headphone board , not yet connected; DIN hardware is not shown. Pico wiring comment , Uno MIDI instance , DAC write .	Use the current inter-board schematic to continuity-check both UART paths, supply terminals, and common ground; measure Pico RX. Build and test the proposed mono line and headphone outputs before treating their diagrams as as-built or quoting audio levels. Select, build, and document the opto-isolated DIN input , then verify its isolation and MIDI reception.

Use the [hardware test log](#) to record observations for each assembled revision.

The older-panel deployment gap was resolved on 2026-09-23: the panel loaded after flashing the Pico W, and its then-current labels reached **Pico: Connected / Uno: Connected to Pico • sound loaded**. The current source shortens the normal Uno label to **Uno: Connected to Pico**; refreshed guide images are simulated, and a new hardware upload has not been performed. Automated browser checks cover controls and keys. The explicit-sync link failure has been reproduced, fixed, and retested on both boards; physical A0 output and per-command Uno reception checks remain open.

Hands-on acceptance pass, when both boards are available

1. Record board/core and library versions, the three source hashes from the [index](#), and the assembled wiring revision.
2. Record the owner-confirmed return-wire level shift, continuity-checked jumper endpoints, and measured Pico RX voltage, then power both boards via their USB ports; check Wi-Fi, `quarkwave.local`, IP fallback, separate **Pico** and **Uno** connection labels, note playback once the audio stage exists, and note release on browser disconnect.

3. Compare browser controls with Uno behavior for each group: envelope, tone, modulation, arpeggiator, delay, chorus, bitcrush, tremolo, drive/fold, and Panic. Log MIDI CC values and compare panel readouts, including vibrato rate and master volume; repeat Randomize and check its full-range effects. Check keyboard velocity 1, 100, and 127 and view switching with a held note.
4. Save and load a user patch, factory preset, factory-to-user copy, and Commit snapshot; restart each board separately and confirm the selected Pico patch is restored on clean Uno startup without opening a browser. Test a lost completion acknowledgment and return-wire disconnection.
5. Exercise DIN notes, bend, CC, and clock without the Pico; then test the Pico RTP gateway with notes, bend, CC, and clock. Confirm standalone activity blocks automatic takeover, while clock alone does not. Check the documented Program Change and SysEx allowlist on each input, including malformed messages and standalone status. Check that BLE is absent in this build.
6. Update the guides with verified results, then remove resolved entries from this queue or link them to development issues.